

Unit 2: HTML

A Deep-dive on the basic fundamentals of HTML

ITM (SLS) Baroda University, Vadodara | Faculty Lecture Series

Department of Computer Science & Engineering

Basics of HTML

An essential guide to understanding the structure of the web

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What is HTML?



The Standard

- Stands for **Hyper Text Markup Language**
- The standard markup language for creating Web pages



Structure & Elements

- Describes the **visual structure** of a Web page
- Consists of a series of nested elements that define layout blocks



Browser Rendering

- Tells the browser exactly **how to display** the content
- Labels pieces of content such as headings, paragraphs, and links

A Simple HTML Document



```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>
  <body>
    <h1>My First Heading</h1>
    <p>My first paragraph.</p>
  </body>
</html>
```

Example Explained

<!DOCTYPE html>

Defines that this document is an HTML5 document.

<html>

The root element of an HTML page.

<head>

Contains meta information about the HTML page.

<title>

Specifies a title for the HTML page (shown in the browser's title bar or in the page's tab).

<body>

Defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.

<h1>

Defines a large heading.

<p>

Defines a paragraph.

What is an HTML Element?

DEFINITION & ANATOMY

An HTML element is defined by a start tag, some content, and an end tag:

```
<p>My first paragraph.</p>
```

Start TagElement ContentEnd Tag

Empty Elements Some elements (like `<br>`) have no content and no end tag.

ELEMENT EXAMPLES

Start TagElement ContentEnd Tag

```
<h1>My First Heading</h1>
```

```
<p>My first paragraph.</p>
```

```
<br>none
```

Start tag	Element content	End tag
<h1>	My First Heading	</h1>
<p>	My first paragraph.	</p>
 	none	none

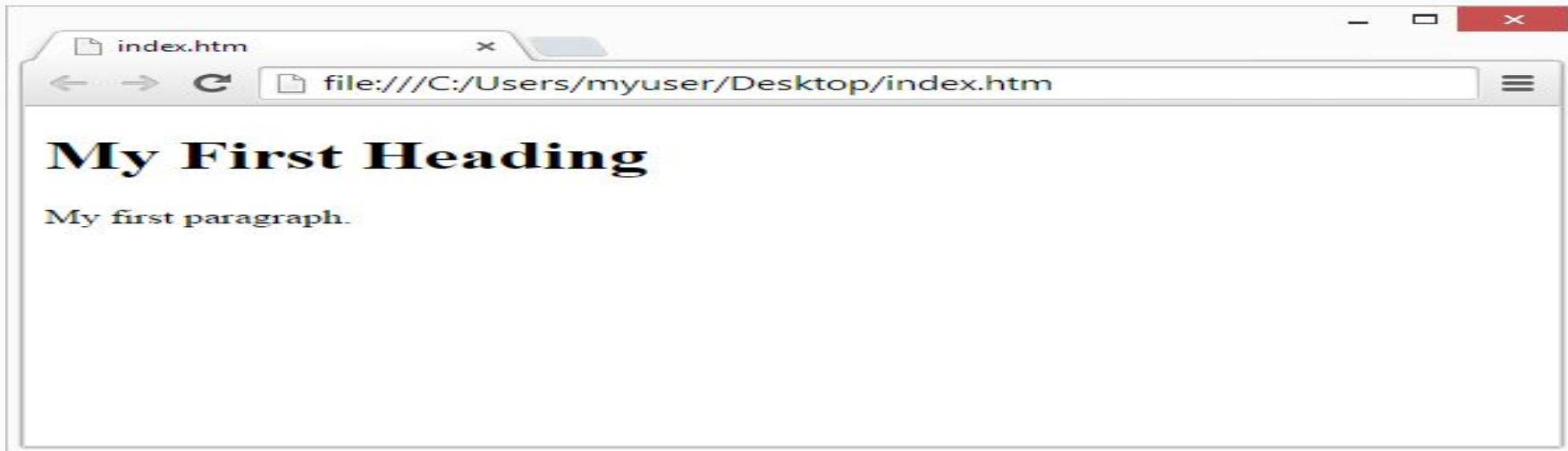
Web Browsers

Core Purpose

To read HTML documents and render them correctly across all modern browsers, including Chrome, Edge, Firefox, and Safari.

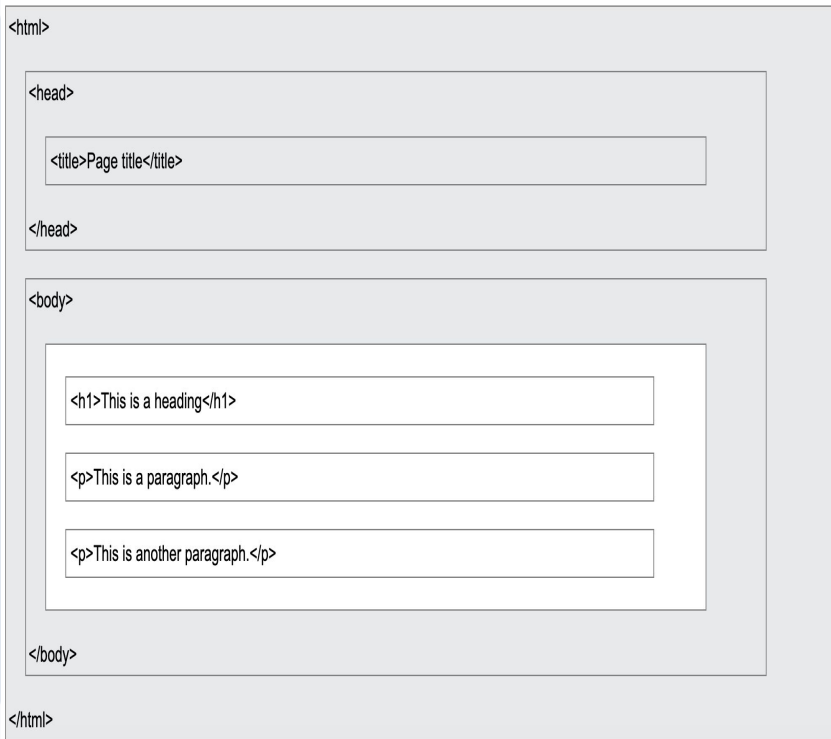
Rendering Mechanism

A browser does not display the raw HTML tags themselves, but instead interprets them as instructions to layout and display the page content.



HTML Page Structure

A visual representation of how elements nest within an HTML document



Document Anatomy

<head> & <title>

The header area contains page metadata. The `<title>` element specifies the text shown in the browser's title bar or tab.

<body>

The visible portion of the document. All content written here—headings, paragraphs, images, and links—renders directly in the main browser window.

HTML History

The evolution of web markup from the inception of the World Wide Web to modern standards

Foundations

1989

Tim Berners-Lee invented WWW

1991

Tim Berners-Lee invented HTML

1993

Dave Raggett drafted HTML+

1995

HTML Working Group defined
HTML 2.0

Standardization

1997

W3C Recommendation: HTML 3.2

1999

W3C Recommendation: HTML 4.01

2000

W3C Recommendation: XHTML 1.0

Modern HTML5

2008 – 2012

WHATWG HTML5 Drafts & Living
Standard

2014

W3C Recommendation: HTML5

2016

W3C Candidate Rec: HTML 5.1

2017

HTML5.1 2nd Edition & HTML5.2
Releases

HTML Documents

The fundamental rules and visual structure of every web page

Structure Rules

- All HTML documents must start with a document type declaration:
`<!DOCTYPE html>`
- The HTML document itself begins with `<html>` and ends with `</html>`.
- The visible part of the document is nested between `<body>` and `</body>`.

```
<!DOCTYPE html>

<html>
  <body>
    <h1>My First Heading</h1>
    <p>My first paragraph.</p>
  </body>
</html>
```

The <!DOCTYPE> Declaration

Informs the web browser about the document type to ensure correct and consistent page rendering

Declaration Rules

- Represents the document type and helps web browsers to display pages correctly.
- Must appear **only once**, at the very top of the page (before any HTML tags).
- The declaration is **not case-sensitive**.
- The modern standard for HTML5 is:
`<!DOCTYPE html>`



```
<!-- Required at the very top -->
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h1>Hello World</h1>
```

```
<p>This is a web page.</p>
```

```
</body>
```

```
</html>
```

HTML Headings

Defining semantic and visual hierarchy to structure web page content

Heading Rules

- HTML headings are defined with the `<h1>` to `<h6>` tags.
- `<h1>` defines the most important heading (highest visual weight).
- `<h6>` defines the least important heading (lowest visual weight).



```
<!-- Headings implementation example  
-->
```

```
<h1>This is heading 1</h1>
```

```
<h2>This is heading 2</h2>
```

```
<h3>This is heading 3</h3>
```

HTML Paragraphs

Defining block-level text elements to structure written content on a web page

Paragraph Rules

- HTML paragraphs are defined with the `<p>` tag.
- They are **block-level elements**, meaning they always start on a new line.
- Browsers automatically add some **white space** (margin) before and after a paragraph.

```
<!-- Paragraphs implementation -->
```

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

HTML Links

Defining hyperlinks to connect web pages and navigate to external resources

Link Rules

- HTML links are defined with the `<a>` tag.
- The link's destination is specified in the `href` attribute.
- Attributes are used to provide additional information about HTML elements.

```
<!-- Links implementation example -->
```

```
<a href="https://www.itmbu.ac.in" >This is a  
link</a>
```

HTML Images

Embedding visual media, defining alternative descriptions, and setting dimensions

Image Rules

- HTML images are defined with the `<img>` tag.
- Attributes like **src** (source), **alt** (alternative text), **width**, and **height** configure the image properties.

```
<!-- Images implementation example -->
```

```

```

How to View HTML Source

Explore, analyze, and experiment with the underlying code of any web page in real-time

View Page Source

Perfect for viewing the entire, original raw HTML document in a static, read-only view.

- Press **CTRL + U** (Windows) or **Cmd + Option + U** (Mac).
- Alternatively, right-click anywhere on the page and select **"View Page Source"**.
- This action opens a brand new browser tab containing the complete source code.



Inspect HTML Elements

Best for examining, styling, and debugging individual elements interactively.

- Right-click directly on any specific element and choose **"Inspect"**.
- The DevTools panel will open, highlighting both the live **HTML structure** and associated CSS styles.
- You can temporarily edit text, swap classes, or change colors *on-the-fly* to test layouts.

HTML Elements

The building blocks of web pages, defining structure from start to end tags

Structure & Rules

- An HTML element is everything from the start tag to the end tag.
- Basic syntax:
`<tagname>Content</tagname>`
- Some elements have no content and no end tag (e.g., `<br>`).
- These are called **empty elements**.

Element Breakdown

Start Tag Content End Tag

`<h1>`My First Heading `</h1>`

`<p>`My first paragraph.`</p>`

`<br>`

Nested HTML Elements

HTML elements can be nested, meaning they can contain other elements

Nesting Principles

- All HTML documents consist of nested HTML elements that build the document structure.
- Nesting means that HTML elements can contain other elements inside them.
- Always ensure tags are closed in the correct reverse order of their opening (e.g., `<html><body>...</body></html>`).

Example Document

```
<!DOCTYPE html>
<html>
  <body>
    <h1>My First Heading</h1>
    <p>My first paragraph.</p>
  </body>
</html>
```

Example Explained

Understanding how HTML elements construct the foundational structure of a web page

The HTML Document Root

- The `<html>` element is the root element and it defines the entire HTML document.
- It strictly consists of a start tag `<html>` and an end tag `</html>`.
- Inside the root container, the `<body>` element encloses all visible page contents.



Visualizing Nesting

```
<html>

  <body>

    <h1>My First Heading</h1>

    <p>My first paragraph.</p>

  </body>

</html>
```

Example Explained (Contd.)

Detailed breakdown of the body, heading, and paragraph elements in HTML

The <body> Element

Defines the document's body. It encloses all visible page content, containing other elements like headings and paragraphs.

```
<body>
  <h1>My Heading</h1>
  <p>My paragraph.</p>
</body>
```

The <h1> Element

Defines a prominent heading. Always pair the start tag and end tag to establish proper semantic hierarchy.

```
<h1>
  My First Heading
</h1>
```

The <p> Element

Defines a standard paragraph of body text. It automatically introduces vertical margins before and after the block.

```
<p>
  My first paragraph.
</p>
```

Never Skip the End Tag

Relying on automatic tag closure can lead to unpredictable layout rendering and validation errors

Browser Tolerance vs. Standards

- Some HTML elements may still display correctly in the browser even if you accidentally omit the closing tag.
- This happens because browser engines attempt to perform automatic error correction on malformed markup.

⚠ CRITICAL WARNING

Never rely on this behavior! Unexpected rendering bugs, style leakage, and accessibility issues will occur if you forget the end tag.



Flawed Code Example

```
<html>
  <body>
    <p>This is a paragraph ⚠ Missing </p>
    <p>This is a paragraph ⚠ Missing </p>
  </body>
</html>
```

Empty HTML Elements

Understanding HTML tags that contain no content and do not require closing tags

What are Empty Elements?

- HTML elements with no content are called empty elements.
- The `<br>` tag defines a line break, and is a classic example of an empty element.
- Empty elements do not have a closing tag (like `</br>`).



Line Break Example

```
<!-- Inside the HTML body -->
```

```
<p>
```

This is a

```
<br> <!-- Empty Line Break -->
```

paragraph with a line break.

```
</p>
```

HTML is Not Case Sensitive

Understanding tag standards, recommendations, and institutional rules

W3C Standards & Behavior

- HTML tags are not case sensitive: `<P>` means the same as `<p>`.
- The official standard does not strictly require lowercase tags.
- W3C strongly **recommends** lowercase in standard HTML.
- W3C **demand**s lowercase for stricter document types like XHTML.

SOCSET Policy

At **School of Computer Science Engineering & Technology (SOCSET)**, we always enforce clean code standards.

✓ PREFERRED PRACTICE

```
<body>
  <p>Correct lowercase</p>
</body>
```

× DISCOURAGED PRACTICE

```
<BODY>
  <P>Avoid uppercase</P>
</BODY>
```

HTML Tag Reference

W3Schools' tag reference contains additional information about these tags and their attributes.

`<html>`

Defines the root of an HTML document

`<body>`

Defines the document's body

`<h1>` to `<h6>`

Defines HTML headings

For a complete list of all available HTML tags, visit our [HTML Tag Reference](#).

HTML Attributes

Key concepts, specifications, and standard syntax rules for element properties

Core Behavior

- All HTML elements can have attributes, which modify default element behavior or styling.
- Attributes provide vital additional information about elements, such as source paths or link destinations.

Syntax Rules

- Attributes are always specified in the **start tag**, never in the end tag.
- Attributes usually come in name/value pairs like: `name="value"`

The href Attribute

Defining hyperlinks and directing navigation paths in HTML documents

Core Purpose

- The **<a>** tag (anchor element) defines a hyperlink, allowing users to navigate from one page to another.
- The **href** attribute is critical—it specifies the destination URL or path of the linked page.

Syntax & Example

A standard anchor element pointing to an external website destination:

```
<a href="https://www.itmbu.ac.in">  
Visit ITMBU  
</a>
```

The src Attribute

Embedding images and managing paths in HTML documents

Core Behavior

The **** tag embeds an image in a page. The **src** attribute defines the image path.

Syntax Example:

```

```

URL Specifications

1. Absolute URL: Links to external sites.

Example: `src="[link removed]"`

⚠️ **Subject to copyright and external deletion.**

2. Relative URL: Hosted within the website.

Example: `src="img_girl.jpg"` or `src="/images/img.jpg"`

✅ **Recommended. Won't break if domain changes.**

The width and height Attributes

Specifying exact dimensions and reserving layout space for images

Core Purpose

- The **width** and **height** attributes specify the dimensions of the image in pixels.
- Defining these attributes prevents page layout flicker and shift while the image is loading.

Syntax & Example

A standard image element with defined pixel width and height:

```

```

The alt Attribute

Providing alternate descriptions for accessibility and fallback display

Core Purpose

- **Fallback Text:** Specifies an alternate description if the image cannot be loaded or displayed.
- **Accessibility:** Crucial for screen readers used by visually impaired users to understand page content.
- **Resilience:** Displays when users have slow connections or if the image source path is broken.

Syntax & Example

The alt text should be descriptive and concise, matching the visual content of the image:

```

```

The style Attribute

Applying inline styles to customize the visual appearance of HTML elements

Core Purpose

- **Inline Styling:** Used to define style rules directly within a single element's tag.
- **Presentation:** Quickly customize visual properties such as colors, fonts, margins, sizes, and layout options.

Syntax & Example

Apply styles using the format `property:value` within the style attribute:

```
<p style="color:red;">This is a red  
paragraph.</p>
```

The lang Attribute

Declaring page language and regional country codes for browsers and search engines

Core Purpose

- **Language Declaration:** Include the lang attribute in the <html> tag to declare the language of the Web page to assist search engines and browsers.
- **Country Codes:** Extend language codes with regional variations (e.g., first two characters for language, last two for country).

Syntax & Examples

Specifying English as the language:

```
<html lang="en">
```

Specifying English (US) as regional variant:

```
<html lang="en-US">
```

The title Attribute

Defining auxiliary information about an HTML element that appears as a hover tooltip

Core Purpose

- **Tooltip Display:** The value of the title attribute is typically displayed as a tooltip popup when the user hovers their mouse pointer over the element.
- **Extra Info:** Used to provide advisory or extra helper information that doesn't need to be constantly visible on the page.

Syntax & Example

Adding a tooltip to a paragraph element:

```
<p title="I'm a tooltip">This is a  
paragraph.</p>
```


Always Use Lowercase Attributes

W3C recommendations and internal coding standards for HTML attribute formatting

W3C Guidelines

- **Specification:** The HTML standard does not strictly require lowercase attribute names. You can write `title` or `TITLE`.
- **W3C Recommendation:** However, W3C strongly recommends lowercase attributes in HTML, and strictly demands them for XHTML document types.

SOCSET Standard

- **Strict Policy:** At SOCSET, we always use lowercase attribute names to maintain clean, readable, and compliant codebases.

Preferred Standard Syntax:

```
<p title="tooltip">
```

Avoid Uppercase Attributes:

```
<p TITLE="tooltip">
```

Always Quote Attribute Values

W3C recommendations and practical syntax requirements for quoting HTML attributes

W3C Recommendations

- **Standard:** The HTML standard does not strictly require quotes around attribute values.
- **Best Practice:** W3C strongly recommends quotes in HTML, and demands them for stricter XHTML.

Good Syntax:

```
<a href="https://www.itmbu.ac.in/html/">
```

Bad Syntax (Avoid):

```
<a href=https://www.itmbu.ac.in/html/>
```

The Space Exception

- **Space Limitation:** Quotes are absolutely necessary when the attribute value contains spaces. Otherwise, it will not display correctly.

Broken Syntax Example:

```
<p title=Description of SOCSET>
```

- **SOCSET Standard:** At SOCSET, we always use quotes around attribute values to ensure clean and robust rendering.

Single or Double Quotes?

Rules and standards for handling nested quote marks inside HTML attributes

Standard Usage

- **Common Practice:** Double quotes around attribute values are the most common in HTML, but single quotes can also be used.

Standard Double Quotes Example:

```
<p title="John Nelson">
```

Standard Single Quotes Example:

```
<p title='John Nelson'>
```

Handling Nested Quotes

- **The Conflict:** When the attribute value itself contains quotes, you must alternate quote styles to avoid breaking syntax.

Single Quotes Nesting Double Quotes:

```
<p title='John "ShotGun" Nelson'>
```

Double Quotes Nesting Single Quotes:

```
<p title="John 'ShotGun' Nelson">
```

HTML Headings

Standard markup tags used to define the hierarchical structure and importance of text content

Heading Hierarchy

- **Tag Range:** HTML headings are defined using the `<h1>` to `<h6>` tags.
- **Importance:** `<h1>` defines the most important (highest level) heading, while `<h6>` defines the least important.

Browser Behavior:

Browsers automatically add some white space (a default margin) before and after any heading element.

Code Examples

Standard syntax for writing headings in HTML:

```
<h1>Heading 1</h1>
```

```
<h2>Heading 2</h2>
```

```
<h3>Heading 3</h3>
```

```
<h4>Heading 4</h4>
```

```
<h5>Heading 5</h5>
```

```
<h6>Heading 6</h6>
```

Headings Are Important

Why and how to structure your web documents using HTML heading tags

Best Practices

- **SEO & Indexing:** Search engines use headings to index the structure and content of your web pages.
- **Readability:** Users skim pages by headings. It is vital to use them to outline clear document structure.
- **Hierarchy:** Use `<h1>` for main titles, followed by `<h2>` for sections, and `<h3>` for sub-sections.

Structure Example

```
<h1>Travel Guide</h1>
<h2>Europe</h2>
<h3>France</h3>
<h3>Italy</h3>
<h2>Asia</h2>
<h3>India</h3>
<h3>Thailand</h3>
```

Tip: Use only one `<h1>` per page for the main topic.

Note: Do not use heading tags solely to style text bold or big.

Bigger Headings

Specifying custom sizes for HTML heading elements using inline CSS style attributes

Font Size Control

- **Custom Sizing:** Every HTML heading tag comes with a pre-configured default browser size. Use inline CSS to customize this.
- **The style Attribute:** Added directly inside the opening tag to define presentation rules.
- **CSS Property:** Uses the `font-size` property followed by value units (pixels, ems, or percentages).

Implementation Code

Syntax structure to modify heading scale:

```
<h1 style="font-size: 60px;">
```

Heading 1

```
</h1>
```

Result: Overrides default styles to render text at exactly 60 pixels tall.

HTML Paragraphs

Defining text structure, block rendering, and margins on web pages

Paragraph Basics

- **Paragraph Tag:** The HTML `<p>` element explicitly defines a structural paragraph on a page.
- **Line Breaks:** Paragraphs are block-level elements, meaning they always start on a fresh, new line.
- **Browser Margins:** Browsers automatically inject standard vertical whitespace before and after each paragraph.

Paragraph Example

How paragraphs are coded in the document body:

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

Note: The paragraph content is wrapped perfectly within opening and closing tags.

HTML Tag Reference

A quick-reference guide to standard document layout and structure elements

`<p>`

Paragraph

Defines a paragraph in the document.

`<hr>`

Horizontal Rule

Defines a thematic change in the content.

`<br>`

Line Break

Inserts a single line break in the layout.

`<pre>`

Preformatted

Defines pre-formatted text elements.

Formatting and Fonts

An essential guide to text styling and typography in HTML

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HTML Display

How web browsers handle spaces, line breaks, and responsive window resizing

Display Rules

- **Unpredictable Output:** You cannot be sure exactly how HTML will be displayed; screen sizes and resized windows create different results.
- **Code Spacing:** Adding extra spaces or empty lines in your HTML code will not change how the content is rendered.
- **Automatic Clean-up:** The browser automatically collapses multiple spaces and discards line breaks when rendering the page.

Whitespace Ignored

Even with extra lines and spaces in the source:

```
<p>  
This paragraph  
contains          a lot of spaces  
and lines, but browser ignores it.  
</p>
```

Result: Renders as a single continuous line, completely ignoring the formatted white space in your editor.

HTML Horizontal Rules

Using the `<hr>` tag to define thematic breaks and separate content in web pages

Horizontal Rule Basics

- **Thematic Break:** The HTML `<hr>` element defines a thematic break in a page, typically displayed as a horizontal line.
- **Separation:** It is most commonly used to visually separate distinct content blocks or define transitions.
- **Empty Tag:** The `<hr>` tag is a self-closing or empty element, meaning it has no companion end tag.

Code Example

How thematic breaks separate markup blocks:

```
<h1>This is heading 1</h1>
<p>This is some text.</p>
<hr>
<h2>This is heading 2</h2>
<p>This is some other text.</p>
<hr>
```

Note: The divider lines separate the layout structure when parsed by browsers.

HTML Line Breaks

How to insert manual line breaks within paragraphs without creating new text blocks

Line Break Basics

- **Line Break:** The HTML `<br>` element defines a manual line break in a web page.
- **No New Paragraph:** Use it to force a new line of text without starting a new paragraph container.
- **Empty Element:** The `<br>` tag is an empty/self-closing element, meaning it has no companion end tag.

Code Example

How manual breaks structure paragraph output:

```
<p>This is<br>  
a paragraph<br>  
with line breaks.</p>
```

Note: This code forces the browser to render the text split into three separate vertical lines while keeping it inside a single logical paragraph container.

The Poem Problem

Understanding why standard HTML paragraph tags collapse spaces and new lines

Why It Happens

- **White-Space Collapsing:** Browsers automatically collapse consecutive spaces, tabs, and line breaks in the HTML source code into a single space.
- **Paragraph Formatting:** The standard `<p>` element is designed for continuous body text, not structural or poetic layout.
- **The Problem:** Writing a poem across multiple lines in your code editor will still display as a single line in the browser.

Source Code vs. Output

HTML Source Code:

```
<p>
My Bonnie lies over the ocean.
My Bonnie lies over the sea.
Oh, bring back my Bonnie to me.
</p>
```

Browser Rendered Output:

My Bonnie lies over the ocean. My Bonnie lies over the sea. Oh, bring back my Bonnie to me.

Solution – The HTML `<pre>` Element

Using the `<pre>` tag to display preformatted text with preserved spaces and line breaks

Preformatted Text Basics

- **Definition:** The HTML `<pre>` element defines preformatted text.
- **Preservation:** It preserves both spaces and manual line breaks exactly as they are written in the source code.
- **Monospace Font:** The text inside a `<pre>` element is automatically displayed in a fixed-width, monospace font.

The `pre` tag preserves both spaces and line breaks:

```
My Bonnie lies over the ocean.  
My Bonnie lies over the sea.  
My Bonnie lies over the ocean.  
Oh, bring back my Bonnie to me.
```

Code Example

How preformatted code maintains structural breaks:

```
<pre>  
My Bonnie lies over the ocean.  
My Bonnie lies over the sea.  
My Bonnie lies over the ocean.  
Oh, bring back my Bonnie to me.  
</pre>
```

Note: Browsers preserve the poetic layout, displaying each verse on its own line exactly as written in the source.

HTML Styles

Using the style attribute to define inline CSS rules for colors, fonts, sizing, and layouts

Style Attribute Basics

- **Inline Styling:** The HTML `style` attribute allows you to apply CSS styling rules directly inside an element tag.
- **Property & Value:** Styles consist of a **property** and a **value** separated by a colon, ending with a semicolon.
- **Direct Override:** Inline styles take precedence over external stylesheets, making them ideal for quick, specific overrides.

Code Example & Output

HTML Code:

```
<p style="color:red;">I am red</p>  
<p style="color:blue;">I am blue</p>  
<p style="font-size:30px;">I am big</p>
```

Rendered Preview:

I am red

I am blue

I am big

The HTML Style Attribute

Syntax and structure of inline CSS styling in HTML elements

Style Basics

- **Inline Styling:** Setting the style of an HTML element is done directly inside the element tag using the style attribute.
- **CSS Declarations:** Each style declaration consists of a property and a value, separated by a colon and ending with a semicolon.
- **Style Precedence:** Inline styles override general styles defined in external stylesheets or style blocks.

Syntax & Blueprint

The standard format of the inline style attribute:

```
<tagname style="property:value;">
```

- **tagname:** The HTML element (e.g. p, h1, body)
- **property:** The CSS property you want to change (e.g. color)
- **value:** The value assigned to the CSS property (e.g. blue)

Background Color

Defining background colors for HTML elements using the inline CSS background-color property

Background Color Basics

- **Element Backgrounds:** The CSS background-color property defines the background color for any HTML element.
- **Target Elements:** It can be applied to page containers like `<body>` or individual text blocks like headings and paragraphs.
- **Value Formats:** Supports standard color names (e.g., `powderblue`), hexadecimal codes, or RGB values.

Code Example & Output

HTML Code:

```
<body style="background-color:powderblue;">  
  <h1>This is a heading</h1>  
  <p>This is a paragraph.</p>  
</body>
```

Rendered Output Preview:

This is a heading

This is a paragraph.

Background Color Example

Applying distinct inline background-color styles to separate HTML elements within a page

Multiple Backgrounds

- **Independent Styling:** HTML allows each individual element to declare its own unique background color via the inline style attribute.
- **Scope:** Unlike setting a background on the `<body>`, targeting specific block elements limits the background to the boundaries of those elements.
- **Visual Hierarchy:** Using contrasting background colors helps partition content blocks and draw attention to structural divisions.

Code Example & Output

HTML Code:

```
<body>  
<h1 style="background-color:powderblue;">This is  
a heading</h1>  
<p style="background-color:tomato;">This is a  
paragraph.</p>  
</body>
```

Rendered Output Preview:

This is a heading

This is a paragraph.

Text Color

Defining text colors for HTML elements using the inline CSS color property

Text Color Basics

- **Inline Coloring:** The HTML `color` property specifies the foreground color of text within any HTML element.
- **Element Scope:** It can be applied inline to headings (`<h1>`), paragraphs (`<p>`), or spans to color specific text ranges.
- **Value Formats:** Supports standard color names (e.g. `blue`, `red`), hexadecimal color values, or RGB/HSL coordinates.

Code Example & Output

HTML Code:

```
<h1 style="color:blue;">This is a heading</h1>  
<p style="color:red;">This is a paragraph.</p>
```

Rendered Output Preview:

This is a heading

This is a paragraph.

Fonts

Defining font families for HTML elements using the inline CSS font-family property

Font Basics

- **Font Family:** The CSS font-family property defines the exact font to be used for rendering an HTML element.
- **Fallbacks:** Best practice is to list multiple font names as fallbacks, ending with a generic family name like sans-serif or monospace.
- **Specific Style:** Inline font selection allows individual text segments to project a tailored mood or aesthetic.

Code Example & Output

HTML Code:

```
<h1 style="font-family:verdana;">This is a heading</h1>
```

```
<p style="font-family:courier;">This is a paragraph.</p>
```

Rendered Output Preview:

This is a heading

This is a paragraph.

Text Size

Defining text sizes for HTML elements using the inline CSS font-size property

Text Size Basics

- **Font Size Property:** The CSS font-size property defines the exact text size for an HTML element.
- **Value Formats:** Supports percentage values (e.g., 300%), absolute units like pixels (px), or relative units (em, rem).
- **Visual Hierarchy:** Scaling text dynamically establishes clear content priority and enhances user readability.

Code Example & Output

HTML Code:

```
<h1 style="font-size:300%;">This is a heading</h1>
```

```
<p style="font-size:160%;">This is a paragraph.</p>
```

Rendered Output Preview:

This is a heading

This is a paragraph.

Text Alignment

Defining horizontal text alignment for HTML elements using the inline CSS `text-align` property

Alignment Basics

- **Property Purpose:** The CSS `text-align` property defines the horizontal text alignment for an HTML element.
- **Standard Values:** Supports values like `center`, `left`, `right`, and `justify` to position text block elements.
- **Center Alignment:** Centering headings and paragraphs helps establish a clean, symmetric visual focus on structured pages.

Code Example & Output

HTML Code:

```
<h1 style="text-align:center;">Centered  
Heading</h1>
```

```
<p style="text-align:center;">Centered  
paragraph.</p>
```

Rendered Output Preview:

Centered Heading

Centered paragraph.

HTML Text Formatting

HTML contains several elements for defining text with a special meaning.

Formatting Tags

- **Bold Text ():** Used to draw attention or highlight keywords without implying extra importance.
- **Italic Text (<i>):** Typically used to indicate idiomatic terms, technical terms, or taxonomic designations.
- **Subscript (<sub>):** Defines subscript text, which appears half a character below the normal line.
- **Superscript (<sup>):** Defines superscript text, which appears half a character above the normal line.

Code Example

HTML Source:

```
<p><b>This text is bold</b></p>  
<p><i>This text is italic</i></p>  
<p>This is<sub> subscript</sub> and  
<sup>superscript</sup></p>
```

Rendered Output Preview:

This text is bold

This text is italic

This is subscript and superscript

HTML Formatting Elements

Special elements designed to style and display text with specific visual or semantic meanings.

Visual Formatting Tags

`<b>` - Bold text

`<i>` - Italic text

`<mark>` - Marked / highlighted text

`<small>` - Smaller text

`<del>` - Deleted text (strikethrough)

`<ins>` - Inserted text (underlined)

Semantic & Positioning Tags

`<strong>` - Important text with strong semantic meaning

`<em>` - Emphasized text (typically italicized)

`<sub>` - Subscript text (shifted slightly below standard baseline)

`<sup>` - Superscript text (shifted slightly above standard baseline)

HTML `<b>` and `<strong>` Elements

Defining bold text styling in HTML documents without conveying extra semantic importance.

Element Concept

- **The `<b>` Tag:** The HTML `<b>` element is used to style text in bold physically.
- **Visual Emphasis:** It draws immediate reader attention or highlights keywords on the page.
- **Semantic Meaning:** Bold styling is applied purely as a visual treatment, without conveying any extra semantic importance or urgency.

Code Example

HTML Source:

```
<!DOCTYPE html><html><body>  
<p>This text is normal.</p>  
<p><b>This text is bold.</b></p>  
</body></html>
```

Rendered Output Preview:

This text is normal.

This text is bold.

HTML `<b>` and `<strong>` Elements (Contd.)

Defining semantic text styling using the HTML `<strong>` element for important content.

Element Concept

- **The `<strong>` Tag:** The HTML `<strong>` element defines text with strong importance.
- **Visual Style:** The content inside a strong element is typically displayed in bold by default.
- **Semantic Value:** Unlike `<b>`, `<strong>` has semantic meaning, indicating that the text has serious importance or urgency.

Code Example

HTML Source:

```
<!DOCTYPE html><html><body>  
<p>This text is normal.</p>  
<p><strong>This text is important!</strong></p>  
</body></html>
```

Rendered Output Preview:

This text is normal.

This text is important!

HTML `<i>` and `<em>` Elements

Defining alternate voice, mood, and semantic emphasis in HTML documents.

HTML `<i>` Element

- **Alternate Voice:** Defines text in an alternate voice or mood. Typically displayed in italics.
- **Usage:** Commonly used for technical terms, phrases from other languages, thoughts, or ship names.

HTML Source:

```
<i>This text is italic</i>
```

Rendered Output Preview:

This text is normal.

This text is italic.

HTML `<em>` Element

- **Emphasized Text:** Defines text with semantic emphasis. Typically displayed in italics by browsers.
- **Screen Readers:** A screen reader will pronounce words inside `<em>` with verbal stress and emphasis.

HTML Source:

```
<em>This text is emphasized</em>
```

Rendered Output Preview:

This text is normal.

This text is emphasized.

HTML <small> Element

Defining smaller text, side comments, and legal print in HTML documents.

Element Concept

- **The <small> Tag:** The HTML <small> element is used to style text one size smaller than the surrounding text.
- **Use Cases:** Commonly used for legal disclaimers, copyright details, terms of use, or side comments.
- **Visual Appeal:** It provides clear content hierarchy by physically reducing text scale without losing basic readability.

Code Example

HTML Source:

```
<p>This is some normal text.</p>  
<p><small>This is some smaller  
text.</small></p>
```

Rendered Output Preview:

This is some normal text.

This is some smaller text.

HTML <mark> Element

Defining marked or highlighted text for visual prominence in HTML documents.

Element Concept

- **The <mark> Tag:** The HTML <mark> element defines text that should be marked or highlighted.
- **Visual Relevance:** Typically used to draw the user's attention to specific words, rendering with a yellow background.
- **Semantic Use:** Ideal for highlighting search results or indicating contextually relevant keywords within a passage.

Code Example

HTML Source:

```
<p>Do not forget to buy <mark>milk</mark>  
today.</p>
```

Rendered Output Preview:

Do not forget to buy **milk** today.

HTML `<del>` Element

Defining deleted or removed text for visual clarity in HTML documents.

Element Concept

- **The `<del>` Tag:** The HTML `<del>` element is used to define text that has been deleted or removed from a document.
- **Visual Relevance:** Browsers will usually render a strike-through line directly through the deleted text.
- **Use Cases:** Perfect for showing edits, revisions, outdated prices, or modifications in collaborative documents.

Code Example

HTML Source:

```
<p>My favorite color is <del>blue</del>  
red.</p>
```

Rendered Output Preview:

My favorite color is ~~blue~~ red.

HTML <ins> Element

Defining inserted or added text for visual clarity in HTML documents.

Element Concept

- **The <ins> Tag:** The HTML <ins> element defines a text that has been inserted into a document.
- **Visual Relevance:** Browsers will usually underline inserted text to visually distinguish it from surrounding content.
- **Semantic Use:** Commonly paired with the tag to clearly demonstrate editorial changes, updates, or corrections.

Code Example

HTML Source:

```
<p>My favorite color is <del>blue</del>  
<ins>red</ins>.</p>
```

Rendered Output Preview:

My favorite color is ~~blue~~ red.

HTML <sub> Element

Defining subscript text for chemical formulas and mathematical notation in HTML documents.

Element Concept

- **The <sub> Tag:** The HTML <sub> element is used to define subscript text in a document.
- **Visual Relevance:** Subscript text appears half a character below the normal line, often rendered in a smaller font.
- **Use Cases:** Commonly used for chemical formulas (like H₂O) or mathematical expressions.

Code Example

HTML Source:

```
<p>This is <sub>subscripted</sub> text.</p>
```

Rendered Output Preview:

This is subscripted text.

HTML `<sup>` Element

Defining superscript text for footnotes, mathematical exponents, and ordinal indicators in HTML documents.

Element Concept

- **The `<sup>` Tag:** The HTML `<sup>` element is used to define superscript text in a document.
- **Visual Relevance:** Superscript text appears half a character above the normal line, often rendered in a smaller font.
- **Use Cases:** Commonly used for footnotes (like WWW[1]), mathematical exponents (like x^2), or ordinal abbreviations.

Code Example

HTML Source:

```
<p>This is <sup>superscripted</sup> text.</p>
```

Rendered Output Preview:

This is ^{superscripted} text.

HTML Text Formatting Elements

Quick reference guide to physical and logical inline styling tags in HTML documents.

`<b>`

Defines bold text

`<sub>`

Defines subscripted text

`<em>`

Defines emphasized text

`<sup>`

Defines superscripted text

`<i>`

Defines text in an alternate voice or mood

`<ins>`

Defines inserted text

`<small>`

Defines smaller text

`<del>`

Defines deleted text

`<strong>`

Defines important text

`<mark>`

Defines marked/highlighted text

HTML Quotation and Citation Elements

Defining block quotations and citations within HTML documents.

Element Concept

- **The <blockquote> Tag:** Defines a section that is quoted from another source.
- **The cite Attribute:** Specifies the URL of the source of the quotation, helping search engines and tools identify its origin.
- **Visual Relevance:** Browsers typically indent the content of a <blockquote> element on both left and right margins.

Code Example

HTML Source:

```
<blockquote cite="http://www.worldwildlife.org">  
For 60 years, WWF has worked to help...  
</blockquote>
```

Rendered Output Preview:

Here is a quote from WWF's website:

For 60 years, WWF has worked to help people and nature thrive. As the world's leading conservation organization, WWF works in nearly 100 countries. At every level, we collaborate with people around the world to develop and deliver innovative solutions that protect communities, wildlife, and the places in which they live.

HTML `<blockquote>` for Quotations

Implementing and styling block-level quotation elements in HTML documents.

Element Concept

- **Key Purpose:** Defines a block section that is quoted directly from an external source or reference.
- **Visual Presentation:** Browsers naturally indent both left and right margins of a `<blockquote>` element.
- **Source Attribution:** Utilize the optional 'cite' attribute to embed the source URL directly in the markup.

Code & Preview

HTML Source Code:

```
<blockquote cite="http://www.worldwildlife.org">For  
60 years, WWF has worked to help...</blockquote>
```

Rendered Output Preview:

Here is a quote from WWF's website:

For 60 years, WWF has worked to help people and nature thrive. As the world's leading conservation organization, WWF works in nearly 100 countries. At every level, we collaborate with people around the world to develop and deliver innovative solutions that protect communities, wildlife, and the places in which they live.

HTML `<q>` for Short Quotations

Defining and displaying inline short quotations with automatic browser formatting.

Element Concept

- **Key Purpose:** Defines a short, inline quotation within a paragraph or text block.
- **Auto-Formatting:** Browsers normally insert quotation marks around the text wrapped in the `<q>` tag.
- **Inline Nature:** Unlike `<blockquote>`, the `<q>` tag is meant for short quotes that do not require paragraph breaks.

Code & Preview

HTML Source Code:

```
<p>WWF's goal is to: <q>Build a future where people  
live in harmony with nature.</q></p>
```

Rendered Output Preview:

Browsers usually insert quotation marks around the `q` element.

WWF's goal is to: “Build a future where people live in harmony with nature.”

HTML `<abbr>` for Abbreviations

Marking up abbreviations to provide useful information to browsers, translation systems, and search engines.

Element Concept

- **Key Purpose:** Defines an abbreviation or an acronym, like "HTML", "CSS", "Mr.", "Dr.", "ASAP", "ATM".
- **Semantic Value:** Marking abbreviations can give useful information to browsers, translation systems and search-engines.
- **Global Title:** Use the global 'title' attribute to show the description for the abbreviation/acronym when you mouse over the element.

Code & Preview

HTML Source Code:

```
<p>The <abbr title="World Health  
Organization">WHO</abbr> was founded in 1948.</p>
```

Rendered Output Preview:

The WHO was founded in 1948.

Marking up abbreviations can give useful information to browsers, translation systems and search-engines.

HTML `<address>` for Contact Information

Defining contact information for the author/owner of a document or an article.

Element Concept

- **Key Purpose:** Defines the contact information for the author/owner of a document or an article.
- **Supported Types:** Can be an email address, URL, physical address, phone number, social media handle, etc.
- **Formatting:** Usually renders in italic. Browsers always add a line break before and after the element.

Code & Preview

HTML Source Code:

```
<address>  
Written by John Doe.<br>  
Visit us at:<br>  
Example.com  
</address>
```

Rendered Output Preview:

The HTML address element defines contact information (author/owner) of a document or article.

*Written by John Doe.
Visit us at:
Example.com
Box 564, Disneyland
USA*

HTML `<cite>` for Work Title

Defining the title of a creative work like books, poems, movies, or paintings in HTML.

Element Concept

- **Key Purpose:** Defines the title of a creative work (e.g. a book, a poem, a song, a movie, a painting, a sculpture, etc.).
- **Exclusion Rule:** A person's name is not the title of a work.
- **Formatting:** The text in the `<cite>` element usually renders in italic.

Code & Preview

HTML Source Code:

```
<p><cite>The Scream</cite> by Edvard Munch. Painted in 1893.</p>
```

Rendered Output Preview:

The HTML cite element defines the title of a work.
Browsers usually display cite elements in italic.



The Scream by Edvard Munch. Painted in 1893.

HTML `<bdo>` for Bi-Directional Override

Overriding the default text direction for bidirectional text layouts.

Element Concept

- **Key Purpose:** Defines a bi-directional override, forcing the enclosed text to render in a specific direction.
- **Mandatory Attribute:** Requires the 'dir' attribute with 'rtl' (right-to-left) or 'ltr' (left-to-right) to function.
- **Common Use:** Crucial for correctly displaying mixed-direction scripts like Arabic, Hebrew, or Persian inside standard LTR pages.

Code & Preview

HTML Source Code:

```
<bdo dir="rtl">  
This text will be written from right to left  
</bdo>
```

Rendered Output Preview:

If your browser supports bi-directional override (bdo), the next line will be written from right to left (rtl):

tfel ot thgir morf nettirw eb lliw enil sihT

HTML Quotation and Citation Elements

A reference guide to semantic elements used for markup of quotes, abbreviations, and citations.

<abbr>

Defines an abbreviation or acronym

<address>

Defines contact information for the author/owner of a document

<bdo>

Defines the text direction for bi-directional overrides

<blockquote>

Defines a section that is quoted from another source

<cite>

Defines the title of a creative work

<q>

Defines a short inline quotation

Color and Hyperlinks

An essential guide to text coloring and anchor elements in HTML

ITM (SLS) Baroda University, Vadodara | Faculty Lecture Series

Department of Computer Science & Engineering

HTML Colors

- HTML colors are specified with predefined color names, or with RGB, HEX, HSL, RGBA, or HSLA values.

Color Names

- In HTML, a color can be specified by using a color name:



- HTML supports 140 standard color names.

HTML Colors

HTML colors are specified with predefined color names, or with RGB, HEX, HSL, RGBA, or HSLA values.

Color Names

In HTML, a color can be specified by using a color name:

HTML supports 140 standard color names.



Background Color

You can set the background color for HTML elements using the style attribute.

Code Example

```
<h1 style="background-color:DodgerBlue;">  
  Hello World  
</h1>  
  
<p style="background-color:Tomato;">  
  Lorem ipsum dolor sit...  
</p>
```

Visual Output

Hello World

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat.

Text Color

You can set the color of text in HTML using the style attribute with color values.

Code Example

```
<h1 style="color:Tomato;">
```

```
Hello World
```

```
</h1>
```

```
<p style="color:DodgerBlue;">
```

```
Lorem ipsum...
```

```
</p>
```

```
<p style="color:MediumSeaGreen;">
```

```
Ut wisi enim...
```

```
</p>
```

Visual Output

Hello World

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat.

Ut wisi enim ad minim veniam, quis nostrud exercitation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat.

Border Color

You can set the color of borders in HTML using the style attribute with color values.

Code Example

```
<h1 style="border:2px solid Tomato;" >  
Hello World  
</h1>
```

```
<h1 style="border:2px solid  
DodgerBlue;" >  
Hello World  
</h1>
```

```
<h1 style="border:2px solid Violet;" >  
Hello World  
</h1>
```

Visual Output



Hello World

Hello World

Hello World

Color Values

In HTML, colors can be specified using RGB, HEX, HSL, and translucent RGBA or HSLA values.

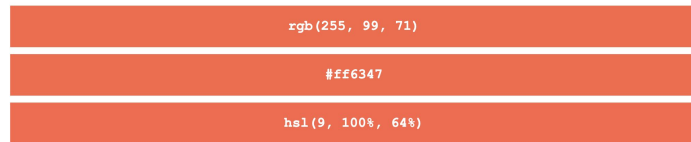
Code Example

```
<h1 style="background-color:rgb(255, 99, 71);">...</h1>
<h1 style="background-color:#ff6347;">...</h1>
<h1 style="background-color:hsl(9, 100%, 64%);">...</h1>

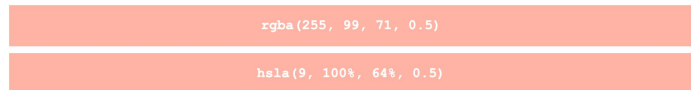
<h1 style="background-color:rgba(255, 99, 71, 0.5);">...</h1>
<h1 style="background-color:hsla(9, 100%, 64%, 0.5);">...</h1>
```

Visual Output

STANDARD COLORS (RGB, HEX, HSL)



WITH ALPHA CHANNEL (50% OPACITY)



HTML RGB and RGBA Colors

The standard color models used in web development to define vibrant light sources and translucent opacities.

RGB Color Model

An RGB color value represents RED, GREEN, and BLUE light sources combined to create various colors.

RED

GREEN

BLUE

RGBA (Alpha Channel)

An RGBA color value is an extension of RGB with an Alpha channel which specifies the opacity of the color.

100%

60%

20%

RGB Color Values

In HTML, colors are defined by combining Red, Green, and Blue light intensities.

The Formula

`rgb(red, green, blue)`

Each parameter defines the light intensity as a value between 0 and 255.

- 256 values per channel
- 256 x 256 x 256 combinations
- **16,777,216 possible colors**

Standard Configurations

`rgb(255, 0, 0)` — RED

`rgb(0, 255, 0)` — GREEN

`rgb(0, 0, 0)` — BLACK

`rgb(255, 255, 255)` — WHITE

RGB Color Values (Contd.)

Examples of applying RGB colors to HTML elements using inline styles.

```
<h1 style="background-color:rgb(255,0,0);">
```

rgb(255, 0, 0)

```
<h1 style="background-color:rgb(0,0,255);">
```

rgb(0, 0, 255)

```
<h1 style="background-color:rgb(60,179,113);">
```

rgb(60, 179, 113)

```
<h1 style="background-color:rgb(238,130,238);">
```

rgb(238, 130, 238)

```
<h1 style="background-color:rgb(255,165,0);">
```

rgb(255, 165, 0)

```
<h1 style="background-color:rgb(106,90,205);">
```

rgb(106, 90, 205)

Shades of Gray

Shades of gray are defined using equal values for all three RGB parameters.

The Gray Rule

`rgb(X, X, X)`

When red, green, and blue values are completely equal, the color saturations cancel out, resulting in a neutral gray.

- Values closer to 0 create darker shades
- Values closer to 255 create lighter shades
- **Total of 256 unique gray levels**

`bg-color: rgb(60, 60, 60)`

rgb(60, 60, 60)

`bg-color: rgb(100, 100, 100)`

rgb(100, 100, 100)

`bg-color: rgb(140, 140, 140)`

rgb(140, 140, 140)

`bg-color: rgb(180, 180, 180)`

rgb(180, 180, 180)

`bg-color: rgb(200, 200, 200)`

rgb(200, 200, 200)

`bg-color: rgb(240, 240, 240)`

rgb(240, 240, 240)

RGBA Color Values

An extension of RGB color values with an Alpha channel to specify color opacity.

The Alpha Parameter

`rgba(R, G, B, alpha)`

The alpha parameter is a number between 0.0 (fully transparent) and 1.0 (fully opaque) that controls transparency levels.

- **0.0** - Completely invisible (transparent)
- **0.5** - Semi-transparent
- **1.0** - Fully opaque (solid color)

`bg-color: rgba(255, 99, 71, 0)`

`rgba(..., 0)`

`bg-color: rgba(255, 99, 71, 0.2)`

`rgba(..., 0.2)`

`bg-color: rgba(255, 99, 71, 0.4)`

`rgba(..., 0.4)`

`bg-color: rgba(255, 99, 71, 0.6)`

`rgba(..., 0.6)`

`bg-color: rgba(255, 99, 71, 0.8)`

`rgba(..., 0.8)`

`bg-color: rgba(255, 99, 71, 1)`

`rgba(..., 1.0)`

HTML HEX Colors

A hexadecimal representation of RGB values used to specify colors in HTML.

The HEX Rule

#RRGGBB

A hexadecimal color is specified with six hexadecimal digits, where RR (red), GG (green), and BB (blue) specify the components.

- **RR (Red):** Specifies red intensity (00 to FF)
- **GG (Green):** Specifies green intensity (00 to FF)
- **BB (Blue):** Specifies blue intensity (00 to FF)

bg-color: **#FF0000**

#FF0000

bg-color: **#00FF00**

#00FF00

bg-color: **#0000FF**

#0000FF

bg-color: **#FFFF00**

#FFFF00

bg-color: **#FFFFFF**

#FFFFFF

bg-color: **#000000**

#000000

HEX Color Values

A hexadecimal representation of RGB values used to specify colors in HTML.

THE HEX RULE

#rrggbb

A hexadecimal color is specified with six digits representing red, green, and blue components between 00 and ff (equivalent to decimal 0–255).

- **rr (Red):** Specifies red intensity (00 to ff)
- **gg (Green):** Specifies green intensity (00 to ff)
- **bb (Blue):** Specifies blue intensity (00 to ff)

bg-color: #ff0000

#ff0000

bg-color: #00ff00

#00ff00

bg-color: #0000ff

#0000ff

bg-color: #ffffff

#ffffff

bg-color: #000000

#000000

HEX Color Values (Contd.)

Examples of using HTML inline styles to define heading background colors.

HTML SOURCE CODE

```
<h1 style="bg-color:#ff0000">
```

```
<h1 style="bg-color:#0000ff">
```

```
<h1 style="bg-color:#3cb371">
```

```
<h1 style="bg-color:#ee82ee">
```

```
<h1 style="bg-color:#ffa500">
```

```
<h1 style="bg-color:#6a5acd">
```

RENDERED OUTPUT

#ff0000

#0000ff

#3cb371

#ee82ee

#ffa500

#6a5acd

Shades of Gray

Shades of gray are defined by using equal values for red, green, and blue parameters.

HTML SOURCE CODE

```
<h1 style="bg-color:#404040">
```

```
<h1 style="bg-color:#686868">
```

```
<h1 style="bg-color:#a0a0a0">
```

```
<h1 style="bg-color:#bebebe">
```

```
<h1 style="bg-color:#dcdcdc">
```

```
<h1 style="bg-color:#f8f8f8">
```

RENDERED OUTPUT

#404040

#686868

#a0a0a0

#bebebe

#dcdcdc

#f8f8f8

HTML HSL and HSLA Colors

Intuitive color models in modern web development offering precision control over lighting and opacity.

Standard Model

HSL Color Model

Represents color values based on three primary visual dimensions:

- **Hue:** Positioned on a 360° color wheel (from red, through green and blue, and back to red).
- **Saturation:** Intensity value from 0% (grayscale shadow) up to 100% (full vibrant color).
- **Lightness:** Brightness level from 0% (pure black) up to 100% (pure white).

Extension Model

HSLA Color Model

Extends the standard HSL color model by introducing opacity control:

- **Alpha Channel:** Specifies the level of transparency for rendering the element.
- **Numeric Range:** Expressed as a decimal between 0.0 (fully transparent) and 1.0 (completely opaque).
- **UI Design:** Ideal for overlay filters, glassmorphism designs, and smooth backdrop blending layers.

HSL Color Values

An intuitive way to specify colors in HTML using Hue, Saturation, and Lightness.

Syntax: `hsl(hue, saturation, lightness)`

01. Parameter

Hue

A degree on the color wheel from 0 to 360.

0 = Red

120 = Green

240 = Blue

02. Parameter

Saturation

A percentage value that represents the color intensity.

0% = Shade of Gray

100% = Full Color

03. Parameter

Lightness

A percentage value that controls the overall brightness.

0% = Pure Black

100% = Pure White

HSL Color Values (Contd.)

Examples of applying HSL colors in HTML style attributes and their rendered results.

```
<h1 style="background-color:hsl(0, 100%, 50%);">...</h1>
```

hsl(0, 100%, 50%)

```
<h1 style="background-color:hsl(240, 100%, 50%);">...</h1>
```

hsl(240, 100%, 50%)

```
<h1 style="background-color:hsl(147, 50%, 47%);">...</h1>
```

hsl(147, 50%, 47%)

```
<h1 style="background-color:hsl(300, 76%, 72%);">...</h1>
```

hsl(300, 76%, 72%)

```
<h1 style="background-color:hsl(39, 100%, 50%);">...</h1>
```

hsl(39, 100%, 50%)

```
<h1 style="background-color:hsl(248, 53%, 58%);">...</h1>
```

hsl(248, 53%, 58%)

Saturation

Saturation can be described as the intensity of a color.



LEVEL 1

100%

Pure color, no shades of gray.



LEVEL 2

50%

50% gray, but you can still see the color.



LEVEL 3

0%

Completely gray; you can no longer see the color.

Saturation (Contd.)

Examples of varying saturation levels of the color red and their rendered results.

```
<h1 style="background-color:hsl(0, 100%, 50%);">...</h1>
```

hsl(0, 100%, 50%)

```
<h1 style="background-color:hsl(0, 80%, 50%);">...</h1>
```

hsl(0, 80%, 50%)

```
<h1 style="background-color:hsl(0, 60%, 50%);">...</h1>
```

hsl(0, 60%, 50%)

```
<h1 style="background-color:hsl(0, 40%, 50%);">...</h1>
```

hsl(0, 40%, 50%)

```
<h1 style="background-color:hsl(0, 20%, 50%);">...</h1>
```

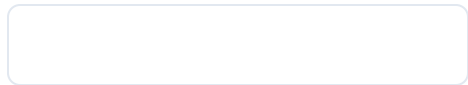
hsl(0, 20%, 50%)

```
<h1 style="background-color:hsl(0, 0%, 50%);">...</h1>
```

hsl(0, 0%, 50%)

Lightness

The lightness of a color can be described as how much light you want to give the color.



LEVEL 1

100%

Full lightness (white).



LEVEL 2

50%

50% light (neither dark nor light).



LEVEL 3

0%

No light (black).

Lightness (Contd.)

Examples of varying lightness levels of the color red and their rendered results.

```
<h1 style="background-color:hsl(0, 100%, 0%);">...</h1>
```

hsl(0, 100%, 0%)

```
<h1 style="background-color:hsl(0, 100%, 25%);">...</h1>
```

hsl(0, 100%, 25%)

```
<h1 style="background-color:hsl(0, 100%, 50%);">...</h1>
```

hsl(0, 100%, 50%)

```
<h1 style="background-color:hsl(0, 100%, 75%);">...</h1>
```

hsl(0, 100%, 75%)

```
<h1 style="background-color:hsl(0, 100%, 90%);">...</h1>
```

hsl(0, 100%, 90%)

```
<h1 style="background-color:hsl(0, 100%, 100%);">...</h1>
```

hsl(0, 100%, 100%)

Shades of Gray

Shades of gray are defined by setting the hue and saturation to 0, and adjusting the lightness from 0% to 100%.

```
<h1 style="background-color:hsl(0, 0%, 20%);">...</h1>
```

hsl(0, 0%, 20%)

```
<h1 style="background-color:hsl(0, 0%, 30%);">...</h1>
```

hsl(0, 0%, 30%)

```
<h1 style="background-color:hsl(0, 0%, 40%);">...</h1>
```

hsl(0, 0%, 40%)

```
<h1 style="background-color:hsl(0, 0%, 60%);">...</h1>
```

hsl(0, 0%, 60%)

```
<h1 style="background-color:hsl(0, 0%, 70%);">...</h1>
```

hsl(0, 0%, 70%)

```
<h1 style="background-color:hsl(0, 0%, 90%);">...</h1>
```

hsl(0, 0%, 90%)

HSLA Color Values

An extension of standard HSL color values, introducing an Alpha channel to specify color opacity.

```
hsla(hue, saturation, lightness, alpha)
```

Hue0 to 360

Represents a degree on the color wheel from 0 to 360. 0 is red, 120 is green, and 240 is blue.

Saturation0% to 100%

A percentage value. 0% means a shade of gray, and 100% is the full, vibrant color.

Lightness0% to 100%

A percentage value. 0% is completely black, 50% is normal (neither dark nor light), and 100% is white.

Alpha0.0 to 1.0

A number specifying opacity. 0.0 is fully transparent, and 1.0 is not transparent at all.

HSLA Color Values (Contd.)

Experiment by mixing the HSLA values below to adjust the alpha (transparency) channel:

```
<h1 style="background-color:hsla(9, 100%, 64%, 0);">...</h1>
```

hsla(9, 100%, 64%, 0)

```
<h1 style="background-color:hsla(9, 100%, 64%, 0.2);">...</h1>
```

hsla(9, 100%, 64%, 0.2)

```
<h1 style="background-color:hsla(9, 100%, 64%, 0.4);">...</h1>
```

hsla(9, 100%, 64%, 0.4)

```
<h1 style="background-color:hsla(9, 100%, 64%, 0.6);">...</h1>
```

hsla(9, 100%, 64%, 0.6)

```
<h1 style="background-color:hsla(9, 100%, 64%, 0.8);">...</h1>
```

hsla(9, 100%, 64%, 0.8)

```
<h1 style="background-color:hsla(9, 100%, 64%, 1);">...</h1>
```

hsla(9, 100%, 64%, 1)

HTML Links – Hyperlinks

The fundamental building blocks of web navigation connecting documents across the internet.



Core Concept

HTML links are hyperlinks. They are found on almost all web pages, allowing users to seamlessly click their way from page to page.



User Interaction

Hovering over a link changes the mouse cursor into a pointer hand. Clicking the element instantly triggers a jump to the target document.



Link Versatility

Links do not have to be plain text. You can easily wrap images, buttons, shapes, or entire HTML layouts inside an anchor element.

HTML Links – Syntax

Understanding the structure, required attributes, and visual representation of anchor elements.

General Structure

The HTML `<a>` tag (anchor) defines a hyperlink. It uses the following syntax:

```
<a href="url">link  
text</a>
```

The `<a>` tag acts as a wrapper, turning any nested content into an active, clickable web gateway.



The href Attribute

Specifies the destination URL of the link. Without it, the anchor tag will not function as a hyperlink.



The Link Text

The visible content displayed on the page. It can consist of plain text, images, or other inline elements.



User Interaction

When clicked, the browser instantly halts the current page view and redirects the user to the specified URL.

HTML Links – Syntax (Contd.)

Understanding default browser styles, anchor states, and output behavior.

Syntax Example

```
<a href="https://itmbu.ac.in">Visit  
itmbu.ac.in!</a>
```

Default Browser Render

HTML Links

[Visit itmbu.ac.in!](https://itmbu.ac.in)

Default Link States



Unvisited Link

[Underlined and blue by default](#)



Visited Link

[Underlined and purple after click](#)



Active Link

[Underlined and red while being active](#)



Pro-tip: These default styles can be easily customized to any design using CSS.

HTML Links – The target Attribute

The target attribute specifies where to open the linked document. By default, it opens in the current window.



_self
DEFAULT

Opens the document in the same window or tab as it was clicked.



_blank
NEW TAB

Opens the document in a brand new window or browser tab.



_parent
FRAMES

Opens the document within the parent frame of the page.



_top
FULL BODY

Opens the document in the full body window, breaking any frames.

HTML Links – The target Attribute (Contd.)

How to configure link behavior to force documents to open in new browser tabs or windows.

Code Example

Applying the `target="_blank"` attribute within the anchor tag:

```
<a href="https://itmbu.ac.in/"  
target=" blank">  
Visit itmbu.ac.in!  
</a>
```

Default Browser Render

Clicking this link launches the destination in a completely new window or tab:

HTML Links

[Visit itmbu.ac.in!](https://itmbu.ac.in/)

Absolute URLs vs. Relative URLs

Understanding how destination paths are defined in HTML anchor tags.



Absolute URLs

FULL WEB ADDRESS

Uses a complete web address (including the protocol) to point to an external site or resource.

```
href="https://www.google.com"
```



Relative URLs

LOCAL LINK

Points to a page or folder within the current website, omitting the protocol and domain parts.

```
href="html_images.asp"
```

Absolute URLs vs. Relative URLs (Contd.)

Comparing the source code structure with the final rendered browser output.

HTML Source Code

The structural markup for both absolute and relative links.

```
<!-- Absolute Links -->
<h2>Absolute URLs</h2>
<p><a href="https://www.w3.org/">W3C</a></p>
<p><a
href="https://google.com/">Google</a></p>

<!-- Relative Links -->
<h2>Relative URLs</h2>
<p><a href="html_images.asp">HTML
Images</a></p>
<p><a href="/css/default.asp">CSS
Tutorial</a></p>
```

Default Browser Render

How standard browsers display these elements on screen.

Absolute URLs

[W3C](https://www.w3.org/)

[Google](https://google.com/)

Relative URLs

[HTML Images](html_images.asp)

[CSS Tutorial](/css/default.asp)

HTML Links – Use an Image as a Link

To use an image as a link, wrap the `<img>` tag inside the `<a>` tag.

HTML Source Code

The structural markup for image links.

```
<!-- Image Link -->
<a href="default.asp">

</a>
```

Default Browser Render

How browsers display the visual link representation.

Image as a Link

The image below is a link. Try to click on it.



Link to an Email Address

Use `mailto:` inside the `href` attribute to create a link that opens the user's email program.

HTML Source Code

The structural markup for email links.

```
<!-- Email Link -->  
<a href="mailto:someone@example.com">  
  Send email  
</a>
```

Default Browser Render

How standard browsers display this link on screen.

To let users send a new email, the link outputs as:

[Send email](mailto:someone@example.com)

Button as a Link

To use an HTML button as a link, you have to add some JavaScript code.

HTML Source Code

JavaScript specifies what happens at certain events.

```
<!-- Button as a Link -->
<button onclick="
  document.location='default.asp'
">
  HTML Tutorial
</button>
```

Default Browser Render

How browsers display the visual button representation.

Button as a Link

Click the button to go to the HTML tutorial.

HTML Tutorial

Link Titles

The title attribute specifies extra information about an element, most often shown as a tooltip on hover.

HTML Source Code

The structural markup for link tooltips.

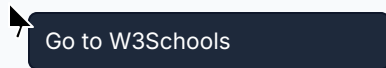
```
<!-- Link with Title -->  
<a href="html_tutorial.asp"  
  title="Go to W3Schools">  
Visit our HTML Tutorial  
</a>
```

Default Browser Render

Hover over the link to see the tooltip message.

Move mouse over the link below:

[Visit our HTML Tutorial](#)



Absolute URLs

An absolute URL uses a full web address to point to an external file or page.

HTML Source Code

The syntax for linking to an external website.

```
<!-- Absolute URL -->  
<a href="[link removed]">  
HTML tutorial  
</a>
```

Default Browser Render

How the browser translates and displays the markup.

External Paths

This example uses a full URL to link to a web page:

[HTML tutorial](#)

Relative URLs (Contd.)

Link to a page located in the html folder on the current web site.

HTML Source Code

The structural markup for folder-relative links.

```
<!-- Relative URL -->  
<a href="/html/default.asp">  
HTML tutorial  
</a>
```

Default Browser Render

How the browser translates and displays the relative link.

External Paths

This example uses a full URL to link to a web page:

[HTML tutorial](#)

Relative URLs (Contd.)

Link to a page located in the html folder on the current web site.

HTML Source Code

The structural markup for folder-relative links.

```
<!-- Relative URL -->  
<a href="/html/default.asp">  
HTML tutorial  
</a>
```

Default Browser Render

How the browser translates and displays the relative link.

External Paths

This example links to a page located in the html folder on the current web site:

[HTML tutorial](/html/default.asp)

Absolute & Relative URLs

Link to a page located in the same folder as the current page.

HTML Source Code

The structural markup for same-folder links.

```
<!-- Same Folder Link -->  
<a href="default.asp">  
HTML tutorial  
</a>
```

Default Browser Render

How the browser translates and displays the link.

External Paths

This example links to a page located in the same folder as the current page:

[HTML tutorial](#)

List, Table, Images, Forms

An essential guide to structured layouts, tabular data, media, and interactive elements in HTML

ITM (SLS) Baroda University, Vadodara | Faculty Lecture Series

Department of Computer Science & Engineering

HTML Lists

HTML lists allow web developers to group a set of related items into organized structures.

Unordered List

The Element

```
<ul>  
<li>Coffee</li>  
<li>Tea</li>  
<li>Milk</li>  
</ul>
```

Browser Output

An Unordered HTML List

- Coffee
- Tea
- Milk

Ordered List

The Element

```
<ol>  
<li>Coffee</li>  
<li>Tea</li>  
<li>Milk</li>  
</ol>
```

Browser Output

An Ordered HTML List

1. Coffee
2. Tea
3. Milk

Unordered HTML List

Defines a bulleted list where the order of items does not explicitly matter. Starts with `<ul>` and uses `<li>` for each item.

HTML Code

The List Structure

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Result Preview

Browser Output

An unordered HTML list

- Coffee
- Tea
- Milk

Ordered HTML List

Defines an ordered list where list items are marked with numbers by default. Starts with `<ol>` and uses `<li>` for each item.

HTML Code

The List Structure

```
<ol>  
  <li>Coffee</li>  
  <li>Tea</li>  
  <li>Milk</li>  
</ol>
```

Result Preview

Browser Output

An Ordered HTML List

1. Coffee
2. Tea
3. Milk

HTML Description Lists

Defines a description list of terms with their corresponding descriptions. Starts with `<dl>`, uses `<dt>` for terms, and `<dd>` for descriptions.

HTML Code

The List Structure

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

Result Preview

Browser Output

A Description List

Coffee
- black hot drink

Milk
- white cold drink

HTML List Tags

A quick reference guide to unordered, ordered, and description list elements in HTML.

USUAL STRUCTURES

Standard Lists

`<ul>`

Defines an unordered (bulleted) list

`<ol>`

Defines an ordered (numbered) list

`<li>`

Defines a list item within a list

KEY-VALUE PAIRS

Description Lists

`<dl>`

Defines a complete description list

`<dt>`

Defines a term within the description list

`<dd>`

Provides the description for the term

HTML Tables

HTML tables allow web developers to arrange data into rows and columns.

Table Structure

The essential tags used to define tabular layouts.

<table>The wrapper element that defines the entire table.

<tr>Defines a table row container for cells.

<th>Defines a header cell (bold and centered by default).

<td>Defines a standard table data cell.

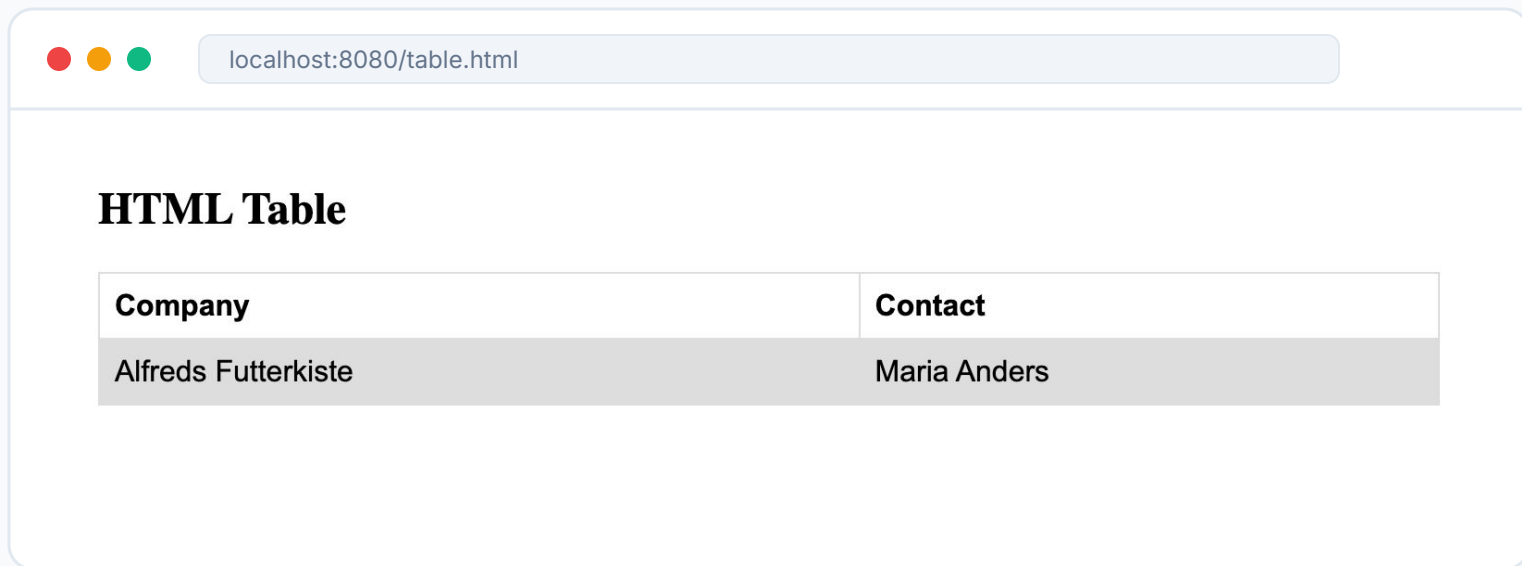
HTML Source Code

The structured markup template for creating tables.

```
<!-- Simple Table structure -->
<table>
<tr>
<th>Company</th>
<th>Contact</th>
</tr>
<tr>
<td>Alfreds Futterkiste</td>
<td>Maria Anders</td>
</tr>
</table>
```

HTML Tables (Contd.)

The visual output rendered by a web browser using the structured HTML table example.



A screenshot of a web browser window. The address bar shows 'localhost:8080/table.html'. The page content includes a heading 'HTML Table' and a table with two columns: 'Company' and 'Contact'. The table has one data row with the values 'Alfreds Futterkiste' and 'Maria Anders'.

Company	Contact
Alfreds Futterkiste	Maria Anders

Define an HTML Table

A table in HTML consists of table cells organized inside systematic rows and columns.

Table Components

Understanding how cells align to build tabular structures.

`<table>`

The primary wrapper container element.

`<tr>`

Defines a table row containing individual cells.

`<th>`

Defines bold and centered header cells (Company, etc).

`<td>`

Defines standard data cells (Alfreds Futterkiste, etc).

HTML Implementation

The structured source code markup for this table.

```
<table>
<tr>
  <th>Company</th>
  <th>Contact</th>
  <th>Country</th>
</tr>
<tr>
  <td>Alfreds Futterkiste</td>
  <td>Maria Anders</td>
  <td>Germany</td>
</tr>
<tr>
  <td>Centro comercial Moctezuma</td>
  <td>Francisco Chang</td>
  <td>Mexico</td>
</tr>
</table>
```

Define an HTML Table (Contd.)

The rendered browser output of the HTML table structure, featuring defined cells with border styling.



localhost:8080/table_borders.html

A basic HTML table

Company	Contact	Country
Alfreds Futterkiste	Maria Anders	Germany
Centro comercial Moctezuma	Francisco Chang	Mexico

To understand the example better, we have added borders to the table.

Table Cells

HTML table cells are defined using the `<td>` element to house and structure diverse data types.

Cell Definition & Rules

The `<td>` Element

- Each table cell is defined by an opening `<td>` and closing `</td>` tag.
- **td** stands for **Table Data**, which holds the actual content.
- Everything between the tags is treated as cell content.

Note: A table cell can contain all sorts of HTML elements: text, images, lists, links, or even nested tables.

Implementation & Preview

```
<!-- Example Code -->
<table>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
</table>
```

Browser Output

TD elements define table cells

Emil	Tobias	Linus
------	--------	-------

To understand the example better, we have added borders to the table.

Table Rows

HTML table rows are defined using the `<tr>` element to group and structure cell data horizontally.

Row Definition & Rules

The `<tr>` Element

- Each table row starts with an opening `<tr>` and ends with a `</tr>` tag.
- **tr** stands for **Table Row**.
- You can have as many rows as you like; just ensure the number of cells is consistent.

Note: There are times when a row can have fewer or more cells than another (covered later).

Implementation & Preview

```
<!-- Example Code -->
<table>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
  <tr>
    <td>16</td>
    <td>14</td>
    <td>10</td>
  </tr>
</table>
```

Browser Output

TR elements define table rows

Emil	Tobias	Linus
16	14	10

To understand the example better, we have added borders to the table.

Table Headers

Use the `<th>` tag instead of the `<td>` tag to define table header cells with bold and centered text by default.

Header Definition & Rules

The `<th>` Element

- Each header cell is defined by an opening `<th>` and closing `</th>` tag.
- **th** stands for **Table Header**.
- By default, browser displays the text in header cells as bold and centered.

Note: Header cells are typically used for the first row or first column of a table.

Implementation & Preview

```
<!-- Example Code -->
<table>
  <tr>
    <th>Person 1</th>
    <th>Person 2</th>
  </tr>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
  </tr>
</table>
```

Browser Output

TH elements define table headers

Person 1	Person 2	Person 3
Emil	Tobias	Linus
16	14	10

To understand the example better, we have added borders to the table.

HTML Table Tags

A reference guide to essential table structure, formatting, and semantic grouping elements.

Core Structure

`<table>` Defines a table

`<th>` Defines a header cell

`<tr>` Defines a table row

`<td>` Defines a table cell

`<caption>` Defines a table caption

Formatting & Groups

`<colgroup>` Groups columns for styling

`<col>` Properties for each column

`<thead>` Groups header content

`<tbody>` Groups body content

`<tfoot>` Groups footer content

HTML Images

Learn how to embed and configure images to enhance the visual design of web pages.

Syntax & Structure

The Element

- Images can improve the design and the appearance of a web page.

HTML Source Code

```
<!-- Syntax Example -->  

```

Preview & Rendering

Browser Output

HTML Image



HTML Images Syntax

A comprehensive guide to image element behavior, key attributes, and syntax structure.

Tag Behavior

How `<img>` Works

- The `<img>` tag is used to embed an image in a web page.
- Images are linked, not directly inserted. The tag creates a holding space.
- The tag is empty, containing attributes only, and has no closing tag.

Required Attributes

src Specifies the path/URL to the image file.

alt Specifies alternate text for screen readers or broken links.

```
<!-- Syntax Example -->  

```

The src Attribute

Understand how web browsers resolve, fetch, and display image paths dynamically.

Attribute Role

How Path Resolution Works

- The required `src` attribute specifies the precise URL or directory path to the target image resource.
- When a page loads, the browser gets the image from the server and inserts it into the viewport instantly.
- Keep image files in consistent locations to prevent broken link icons from displaying alongside the alt text.

Preview & Code

Example Implementation

```
<!-- Syntax Example -->  

```



The alt Attribute

Ensure accessibility and graceful degradation when images fail to render.

Attribute Purpose

Why Alternate Text Matters

- The required **alt** attribute provides fallback text when an image cannot be viewed.
- Essential for users with slow connections, broken image sources, or assistive technologies.
- The value of the attribute should descriptively and accurately convey the image content.

Implementation

Syntax & Visual Output

```
<!-- Syntax Example -->  

```

Alternative text

The alt attribute should reflect the image content, so users who cannot see the image get an understanding of what the image contains:



Image Size – Width and Height

Learn how to specify image dimensions using styles or attributes to prevent layout shifts.

Sizing Methods

How to Define Dimensions

- The `style` attribute defines inline CSS dimensions (e.g., width & height).
- Alternatively, native `width` and `height` attributes define size in pixels.
- **Best Practice:** Always specify size. Without it, pages may flicker or shift layouts while images load.

Implementation

Syntax Examples

```
<!-- 1. Using CSS Style Attribute -->  

```

```
<!-- 2. Using HTML Attributes -->  

```

Image as a Link

- To use an image as a link, put the `<img>` tag inside the `<a>` tag:

Example:

```
<a href="default.asp">
```

```
  
```

```
</a>
```

Image as a Link

Learn how to combine anchor tags and image elements to build visual, clickable hyperlinks.

HTML Structure

Nesting Image Elements

- To use an image as a link, simply place the `<img>` tag inside the `<a>` tag.
- The anchor tag's `href` attribute defines the destination URL.
- The entire image area becomes clickable, inheriting the link properties.

Implementation

Syntax Example

```
<!-- Nested Link Structure -->  
<a href="default.asp">  
    
</a>
```


Common Image Formats

Overview of standard web graphic formats, file extensions, and specifications.

APNG

Animated Portable Network Graphics

`.apng`

GIF

Graphics Interchange Format

`.gif`

ICO

Microsoft Icon format

`.ico, .cur`

JPEG

Joint Photographic Expert Group image

`.jpg, .jpeg, .jfif, .jpp`

PNG

Portable Network Graphics

`.png`

SVG

Scalable Vector Graphics

`.svg`

HTML Image Tags

Essential elements for embedding, mapping, and structuring visual media on the web.

`<img>`

Defines an image to be embedded in the document.

`<map>`

Defines an image map containing interactive clickable areas.

`<area>`

Defines a specific, clickable geometric region inside an image map.

`<picture>`

Defines a responsive container that holds multiple image resources for different viewports.

HTML Forms

An HTML form is used to collect user input, which is most often sent to a server for processing.

HTML Source Code

```
<form action="/action_page.php">
<label for="fname">First name:</label><br>
<input type="text" id="fname" value="John"><br>
<label for="lname">Last name:</label><br>
<input type="text" id="lname"
value="Doe"><br><br>
<input type="submit" value="Submit">
</form>
```

Rendered Form Output

HTML Forms

First name:

Last name:

The <form> Element

The primary container used to structure and gather user inputs.

HTML Form Syntax

```
<form>
  ...
  form elements
  ...
</form>
```

Input Container

The <form> element is a dedicated container designed to house and coordinate multiple user input components, including:

- **Text Fields & Areas** for descriptive text input
- **Checkboxes & Radios** for selection options
- **Submit Buttons** to process and dispatch data

The <input> Element

The most versatile and frequently used component within HTML forms.

Dynamic & Adaptable

An <input> element can be displayed in many ways, depending on the value of its **type** attribute.

By varying this single attribute, developers can capture text, toggle choices, or trigger actions natively.

COMMON TYPE ATTRIBUTES

type="text"

Displays a single-line text input field

type="radio"

Displays a radio button for selecting one option

type="checkbox"

Displays a checkbox for multiple choices

type="submit"

Displays a button to submit form data to a server

Text Fields

The `<input type="text">` defines a single-line input field for user text input.

HTML FORM SYNTAX

```
<form>
  <label for="fname">First
  name:</label><br>
  <input type="text" id="fname"
  name="fname"><br>
  <label for="lname">Last
  name:</label><br>
  <input type="text" id="lname"
  name="lname">
</form>
```

Browser Output

First name:

Last name:

Note: The form container itself is completely invisible.

Width: The default width of an input field is 20 characters.

The <label> Element

Enhancing web accessibility, user experience, and semantic form connections.



Screen Readers

A screen-reader will read the label out loud when a user focuses on the input element, making forms highly accessible.



Easy Interaction

Clicking the text label automatically toggles its associated checkbox or radio button, offering a much larger click region.



Explicit Binding

The `for` attribute of the tag must match the `id` of the target input element to bind them together.

Radio Buttons

The `<input type="radio">` defines a radio button, allowing selection of ONE option from a set.

HTML FORM SYNTAX

```
<form>

  <input type="radio" id="html"
        name="fav_lang" value="HTML">
  <label for="html">HTML</label>

  <input type="radio" id="css"
        name="fav_lang" value="CSS">
  <label for="css">CSS</label>

</form>
```

Browser Output

Choose your favorite Web language:

- ☐ HTML
- ☐ CSS
- ☒ JavaScript

Note: Users can only select one radio option within the same group (matching 'name' attribute).

Tip: Clicking the label text also triggers the selection.

Checkboxes

The `<input type="checkbox">` defines a checkbox, letting a user select ZERO or MORE options.

HTML FORM SYNTAX

```
<form>

  <input type="checkbox" id="v1"
    name="vehicle1" value="Bike">
  <label for="v1">I have a bike</label>

  <input type="checkbox" id="v2"
    name="vehicle2" value="Car">
  <label for="v2">I have a car</label>

</form>
```

Browser Output



I have a bike



I have a car



I have a boat

Note: Checkboxes allow independent selection. Clicking one does not deselect others.

Tip: Wrap text in a `<label>` to make the text itself clickable.

The Submit Button

The `<input type="submit">` defines a button for submitting the form data to a form-handler.

HTML FORM SYNTAX

```
<form action="/action_page.php">
  <label for="fname">First name:</label>
  <input type="text" id="fname"
    value="John">
  <label for="lname">Last name:</label>
  <input type="text" id="lname"
    value="Doe">

  <input type="submit" value="Submit">

</form>
```

Browser Output

First name:

Last name:

Note: The form-handler is typically a file on the server with a script for processing input data.

Tip: The form-handler is specified in the form's action attribute.

HTML5 New Elements

An essential guide to semantic structure, multimedia, and graphic elements in HTML5

ITM (SLS) Baroda University, Vadodara | Faculty Lecture Series

Department of Computer Science & Engineering

HTML5 New Elements

An essential upgrade introducing semantic elements that clearly describe their meaning.

WHAT ARE SEMANTIC TAGS?

Before HTML5, structures relied on generic `<div>` tags (e.g., `id="header"`). HTML5 introduced meaningful tags that describe their purpose clearly to both developers and browsers.

STRUCTURAL EXAMPLES

- `<header>` Page top header
- `<article>` Independent content block
- `<table>` Tabular data representation
- `<footer>` Page bottom section

KEY BENEFITS

Search Engine Optimization (SEO) Helps search engines and crawlers index and understand your page layout and content structure efficiently.

Accessibility (A11y) Enables screen readers to easily navigate and translate the structure for visually impaired users.

Code Readability Improves the developer experience with clean, intuitive, and highly organized source code structure.

HTML5 New Elements

Essential semantic elements for structuring modern, accessible web layouts.

Layout Semantic Tags

Use these standardized elements to organize content areas clearly. This improves structural meaning, making it easier for screen readers and search crawlers to navigate your page.

<header> / <footer>

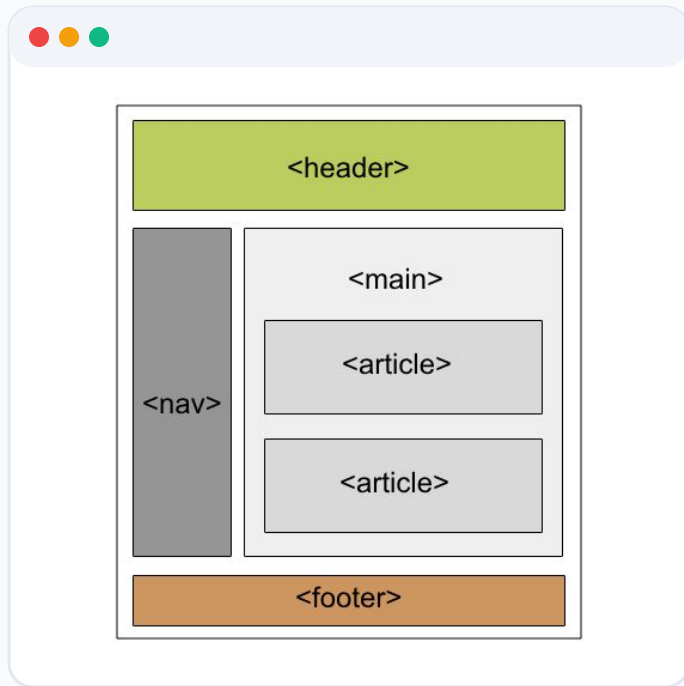
Defines the introduction and bottom metadata of the page.

<nav>

Declares a block of navigation links for website traversal.

<main> / <article>

Identifies primary content and independent, self-contained articles.



HTML5 Semantic Elements (Contd.)

A continuation of key structural and content tags essential for modern web development.

PAGE STRUCTURE

`<header>`

Introductory content or navigational links at the top of a page or section.

`<nav>`

Defines the main navigation menu block (menus, tables of contents).

`<main>`

Wraps the dominant, central content of the document.

`<footer>`

Page bottom containing copyrights, author info, or secondary links.

CONTENT & SECTIONS

`<section>`

A thematic grouping of content, typically starting with its own heading (`<h1>` - `<h6>`).

`<article>`

Represents independent, self-contained content (e.g., a blog post or news story).

`<aside>`

Content indirectly related to the main content, often placed as a sidebar (e.g., related links, ads).

HTML5: Drag/Drop, Video, Audio, Input types

An essential guide to native rich media, interactive drag-and-drop, and advanced form inputs in HTML5

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HTML5: Multimedia, Input Types, and APIs

Native support for rich media, eliminating the need for legacy browser plugins.

NATIVE BENEFITS

Standardized Playback

HTML5 introduces first-class media elements that run natively inside browsers without external software.

✓ No Third-Party Plugins

Eliminates security risks and overhead associated with tools like Adobe Flash.

✓ Direct Page Embedding

Embed media assets directly into the markup for better performance and DOM control.

IMPLEMENTATION

The <video> Element

```
<video width="600" controls autoplay loop
muted>
<source src="lecture.mp4" type="video/mp4">
Your browser does not support the video tag.
</video>
```


HTML5: Multimedia, Input Types, and APIs (Contd.)

The <audio> element brings native, plugin-free audio streaming directly to the web browser.

IMPLEMENTATION

The <audio> Element

```
<audio controls>
  <source src="podcast.mp3"
  type="audio/mpeg">
</audio>
```

Specify multiple sources using the nested source tag to support different audio formats like MP3, WAV, or OGG across various browsers.

CONTROL & BEHAVIOR

Key Attributes

controls

Adds essential user play, pause, and volume controls to the interface.

autoplay

Instructs the browser to start playing the audio file automatically.

loop

Restarts the audio sequence automatically from the beginning when done.

muted

Loads and begins playing the audio with the volume initially turned off.

New Form Input Types

HTML5 forms introduce built-in validation and native mobile keyboards

Smarter Forms

Native Browser Power

HTML5 made forms much smarter. Browsers now provide built-in validation and specialized mobile keyboards without needing complex JavaScript.

email `<input type="email">`

Validates that the input contains an @ symbol.

date `<input type="date">`

Opens a native pop-up calendar to pick a date.

color `<input type="color">`

Opens the operating system's native color picker.

range `<input type="range">`

Renders as a draggable slider for selecting values.

number `<input type="number">`

Restricts typing to digits and adds stepper arrows.

The Drag and Drop API

HTML5 allows natively dragging elements using simple attributes and standard JavaScript events.

NATIVE ATTRIBUTE

Making Elements Draggable

By default, only images, links, and text selections are draggable. To make any other HTML element draggable, add the native attribute:

```
draggable="true"
```

✓ Applies to standard containers like `<div>`, `<p>`, and custom elements.

JAVASCRIPT EVENTS

Key Drag & Drop Events

`ondragstart`

Triggers immediately when the user clicks and begins dragging the element.

`ondragover`

Specifies where dragged elements can be dropped. Must prevent default browser blocking to allow drops.

`ondrop`

Executes custom drop logic when the draggable element is finally released.

XHTML: Meta Tags

An essential guide to document metadata, indexing, and character encoding in XHTML

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XHTML and Meta Tags

While HTML5 is forgiving (it doesn't crash if you forget to close a tag), XHTML is the strict, XML-based version of HTML.

STRICT STANDARD

Predictable XML Parsing

Unlike HTML5, XHTML does not tolerate syntax shortcuts. By enforcing XML conformance, it ensures your code remains consistent, clean, and easily parsed by automated tools and enterprise systems.

The Core Rules of XHTML

1. Strictly Lowercase Tags

All tags must be lowercase. Use `<body>`, never `<BODY>`.

2. Properly Closed Elements

Empty elements must include a closing slash: `<br />` instead of `<br>`.

3. Perfect Nesting Hierarchy

Tags must be nested in precise order: `<b><i>bold italic</i></b>`.

Meta Tags

Provide essential metadata to browsers, search engines, and web services inside the <head> element.

DOCUMENT HEAD

Invisible Metadata

Meta tags reside exclusively in the HTML header. While invisible to visitors, they communicate critical configuration, SEO, and viewport behavior to machines.

```
<head>
  <meta ...>
  <title>My Page</title>
</head>
```

charset <meta charset="UTF-8">

Defines character encoding (ensures emojis & accents render correctly).

viewport <meta name="viewport" content="...">

Enables responsive web design by scaling pages for mobile screens.

description <meta name="description" content="...">

Provides page summary used directly by Google in search results.

author <meta name="author" content="Your Name">

Specifies the author of the document.

refresh <meta http-equiv="refresh" content="30">

Refreshes the page automatically every 30 seconds.